The documentation and process conversion measures necessary to comply with this document shall be completed by 26 June 2024.

INCH-POUND
MIL-PRF-19500/357P
w/AMENDMENT 1
26 March 2024
SUPERSEDING
MIL-PRF-19500/357P
27 August 2020

PERFORMANCE SPECIFICATION SHEET

# TRANSISTOR, PNP, RADIATION HARDENED, SILICON, AMPLIFIER, TYPES 2N3634 THROUGH 2N3637, JAN, JANTX, JANTXV, JANS, JANHC AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

## 1. SCOPE

* 1.1 Scope. This specification covers the performance requirements for PNP, silicon, radiation hardened, low-power amplifier, and switching transistors. Four levels of product assurance are provided for each encapsulated device (JAN, JANTX, JANTXV, and JANS) and two levels of product assurance are provided for unencapsulated die (JANHC and JANKC). Provisions for RHA to eleven radiation levels are provided for JANS and JANKC. RHA level designators “E”, “K”, “U”, “M”, “D”, “P“, “L”, “R”, “F’, “G”, and “H” are appended to the device prefix to identify devices, which have passed RHA requirements.

1.2 Package outlines and die topography. The device package for the encapsulated device type are as follows: TO-5 and TO-39 in accordance with figure 1, surface mount version in accordance with figure 2. The dimensions and topography for JANHC and JANKC unencapsulated die are as follows: “A” version die in accordance with figure 3, “B” version die in accordance with figure 4, and “C” version die in accordance with figure 5.

1.3 Maximum ratings. Unless otherwise specified $T_A$ = +25°C.

| Types | $P_T$ (1) $T_A$ = +25°C | $P_T$ (2) $T_C$ = +25°C | $P_T$ (3) $T_{SP}$ = +25°C | $R_{\theta JA}$ (4) | $R_{\theta JC}$ (4) | $R_{\theta JSP}$ (4) | $I_C$ | $T_J$ and $T_{STG}$ | $V_{CBO}$ | $V_{CEO}$ | $V_{EBO}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | W | W | W | °C/W | °C/W | °C/W | A dc | °C | V dc | V dc | V dc |
| 2N3634, 2N3634L | 1 | 5 | N/A | 175 | 35 | N/A | 1 | -65 to +200 | -140 | -140 | -5 |
| 2N3634UB and UBN | 0.5 | N/A | 1.5 | 325 | N/A | 90 | 1 | | -140 | -140 | -5 |
| 2N3635, 2N3635L | 1 | 5 | N/A | 175 | 35 | N/A | 1 | | -140 | -140 | -5 |
| 2N3635UB and UBN | 0.5 | N/A | 1.5 | 325 | N/A | 90 | 1 | | -140 | -140 | -5 |
| 2N3636, 2N3636L | 1 | 5 | N/A | 175 | 35 | N/A | 1 | | -175 | -175 | -5 |
| 2N3636UB and UBN | 0.5 | N/A | 1.5 | 325 | N/A | 90 | 1 | | -175 | -175 | -5 |
| 2N3637, 2N3637L | 1 | 5 | N/A | 175 | 35 | N/A | 1 | | -175 | -175 | -5 |
| 2N3637UB and UBN | 0.5 | N/A | 1.5 | 325 | N/A | 90 | 1 | | -175 | -175 | -5 |

(1) See figure 6 and figure 7.
(2) See figure 8.
(3) See figure 9.
(4) See figures 10, 11, and 12.

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* 1.4 Primary electrical characteristics. Unless otherwise specified, $T_A$ = +25°C.

| Types | $h_{FE}$ at $V_{CE}$ = -10 V dc (1) | | | | | | | $\lvert h_{fe} \rvert$ $V_{CE}$ = -30 V dc $I_C$ = 30 mA dc f = 100 Mhz | | $C_{obo}$ $V_{CB}$ = -20 V dc $I_E$ = 0 100 Khz ≤ f ≤ 1 Mhz |
|---|---|---|---|---|---|---|---|---|---|---|
| | $h_{FE1}$ $I_C$ = 0.1 mA dc | $h_{FE2}$ $I_C$ = 1.0 mA dc | $h_{FE3}$ $I_C$ = 10 mA dc | $h_{FE4}$ $I_C$ = 50 mA dc | | $h_{FE5}$ $I_C$ = 150 mA dc | | | | |
| | Min | Min | Min | Min | Max | Min | Max | Min | Max | Max |
| 2N3634, 2N3634L | 25 | 45 | 50 | 50 | 150 | 30 | | 1.5 | 8.0 | 10 |
| 2N3634UB and UBN | 25 | 45 | 50 | 50 | 150 | 30 | | 1.5 | 8.0 | 10 |
| 2N3635, 2N3635L | 55 | 90 | 100 | 100 | 300 | 60 | | 2.0 | 8.5 | 10 |
| 2N3635UB and UBN | 55 | 90 | 100 | 100 | 300 | 60 | | 2.0 | 8.5 | 10 |
| 2N3636, 2N3636L | 25 | 45 | 50 | 50 | 150 | 30 | | 1.5 | 8.0 | 10 |
| 2N3636UB and UBN | 25 | 45 | 50 | 50 | 150 | 30 | | 1.5 | 8.0 | 10 |
| 2N3637, 2N3637L | 55 | 90 | 100 | 100 | 300 | 60 | | 2.0 | 8.5 | 10 |
| 2N3637UB and UBN | 55 | 90 | 100 | 100 | 300 | 60 | | 2.0 | 8.5 | 10 |

| | $V_{CE(sat)1}$ $I_C$ = -10 mA dc (1) $I_B$ = -1 mA dc | $V_{CE(sat)2}$ $I_C$ = -50 mA dc (1) $I_B$ = -5 mA dc | $V_{BE(sat)1}$ $I_C$ = -10 mA dc (1) $I_B$ = -1 mA dc | $V_{BE(sat)2}$ $I_C$ = -50 mA dc (1) $I_B$ = -5 mA dc | Switching parameters | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | $t_d$ | $t_r$ | $t_s$ | $t_f$ |
| | V dc | V dc | V dc | V dc | ns | ns | ns | ns |
| Minimum | | | | -0.65 | | | | |
| * Maximum | -0.3 | -0.6 | -0.8 | -0.90 | 100 | 100 | 500 | 150 |

(1) Pulsed (see 4.5.1).

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.5 for PIN construction example and 6.6 for a list of available PINs.

1.5.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: “JAN”, “JANTX”, “JANTXV” and "JANS".

1.5.2 JAN certification mark and quality level for unencapsulated devices (die). The quality level designators for unencapsulated devices (die) that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

* 1.5.3 Radiation hardness assurance (RHA) designator. The RHA levels that are applicable for this specification sheet from lowest to highest are as follows: “E”, “K”, “U”, "M", "D", "P", "L", "R", "F", "G", and "H".

1.5.4 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.4.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

1.5.4.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "3634", "3635", “3636”, and "3637".

1.5.4.3 Suffix symbols. The suffix letter "L" is used on devices that are packaged in the TO-5 package of figure 1 that have a long lead length: 1.500 inches (38.10 mm) minimum and 1.750 inches (44.45 mm) maximum. Devices with standard length leads (see figure 1, TO-39) use no suffix. The suffix letters “UB” or “UBN” are used on devices that are packaged in the surface mount package of figure 2. The “UBN” suffix has an isolated metal lid where the “UB” suffix does not.



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1.5.5 Die identifiers for unencapsulated devices (manufacturers and critical interface identifiers). The manufacturer die identifiers that are applicable for this specification sheet are "A", “B”, and "C" and are in figures 3, 4, and 5 respectively.

1.5.6 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

* 1.6 Radiation features. The following radiation features are applicable for RHA devices supplied to this specification sheet.

* 1.6.1 Maximum total ionizing dose (TID). The maximum TID that RHA devices were tested to in accordance with condition A (dose rate = 50 to 300 rad(Si)/s) of method 1019 of MIL-STD-750 are as follows:

For device type 2N3634, 2N3634L, 2N3634UB, 2N3634UBN: ..... 300 krads(Si) 1/

For device type 2N3635, 2N3635L, 2N53635UB, 2N3635UBN: ..... 300 krads(Si) 1/

For device type 2N3636, 2N3636L, 2N53636UB, 2N3636UBN: ..... 300 krads(Si) 1/

For device type 2N3637, 2N3637L, 2N53637UB, 2N3637UBN: ..... 300 krads(Si) 1/

* 1.6.2 Maximum total ionizing dose (TID). The maximum TID that RHA devices were tested to in accordance with condition D (dose rate ≤ 10 mrad(Si)/s) of method 1019 of MIL-STD-750 are as follows:

For device type 2N3634, 2N3634L, 2N3634UB, 2N3634UBN: ..... 100 krads(Si) 2/

For device type 2N3635, 2N3635L, 2N53635UB, 2N3635UBN: ..... 100 krads(Si) 2/

For device type 2N3636, 2N3636L, 2N53636UB, 2N3636UBN: ..... 100 krads(Si) 2/

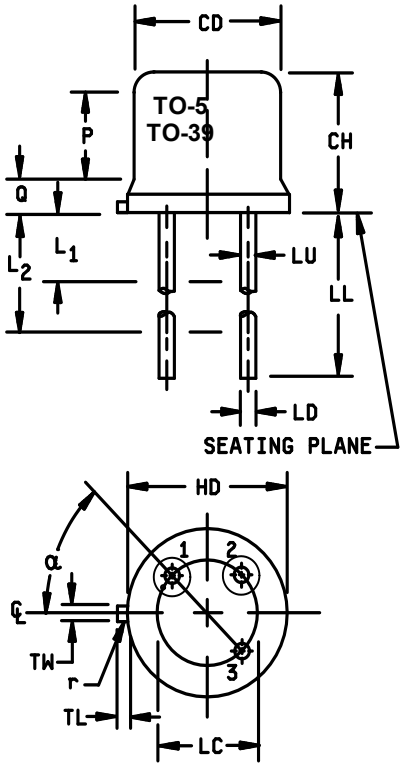
For device type 2N3637, 2N3637L, 2N53637UB, 2N3637UBN: ..... 100 krads(Si) 2/

* 1/ The manufacturers supplying these device types have performed characterization testing in accordance with condition A (dose rate = 50 to 300 rad(Si)/s) of method 1019 of MIL-STD-750. The radiation end point limits are guaranteed only to a maximum TID level of 300 krads(Si).
* 2/ The manufacturers supplying these device types have performed characterization testing in accordance with condition D (dose rate ≤ 10 mrad(Si)/s) of method 1019 of MIL-STD-750. The radiation end point limits are guaranteed only to a maximum TID level of 100 krads(Si).



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| Ltr | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| CD | .305 | .335 | 7.75 | 8.51 | |
| CH | .240 | .260 | 6.10 | 6.60 | |
| HD | .335 | .370 | 8.51 | 9.40 | |
| LC | .200 TYP | | 5.08 TYP | | 7 |
| LD | .016 | .021 | 0.41 | 0.53 | 6 |
| LL | See notes 7, 9, and 10 | | | | |
| LU | .016 | .019 | 0.41 | 0.48 | 7 |
| $L_1$ | | .050 | | 1.27 | 7 |
| $L_2$ | .250 | | 6.35 | | 7 |
| P | .100 | | 2.54 | | 5 |
| Q | | .050 | | 1.27 | |
| r | | .010 | | 0.254 | 8 |
| TL | .029 | .045 | 0.74 | 1.14 | 4 |
| TW | .028 | .034 | 0.71 | 0.86 | 3 |
| $\alpha$ | 45° TP | | 45° TP | | 6 |
| Term 1 | Emitter | | | | |
| Term 2 | Base | | | | |
| Term 3 | Collector | | | | |




NOTES:

1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Beyond r maximum, TW must be held to a minimum length of .021 inch (0.53 mm).
4. TL measured from maximum HD.
5. CD shall not vary more than ±.010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
6. Leads at gauge plane .054 - .055 inch (1.37 - 1.40 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at a maximum material condition (MMC) relative to the tab at MMC. The device may be measured by direct methods or by gauge and gauging procedure.
* 7. LU applies between $L_1$ and $L_2$. LD applies between $L_2$ and LL minimum. Diameter is uncontrolled in $L_1$ and beyond LL minimum.
8. r (radius) applies to both inside corners of tab.
9. For transistor types 2N3634 through 2N3637, LL is .500 inch (12.70 mm) minimum, and .750 inch (19.05 mm) maximum (TO-39).
10. For transistor types 2N3634L through 2N3637L, LL is 1.500 inches (38.10 mm) minimum, and 1.750 inches (44.45 mm) maximum (TO-5).

* FIGURE 1. Physical dimensions (TO-5 and TO-39).



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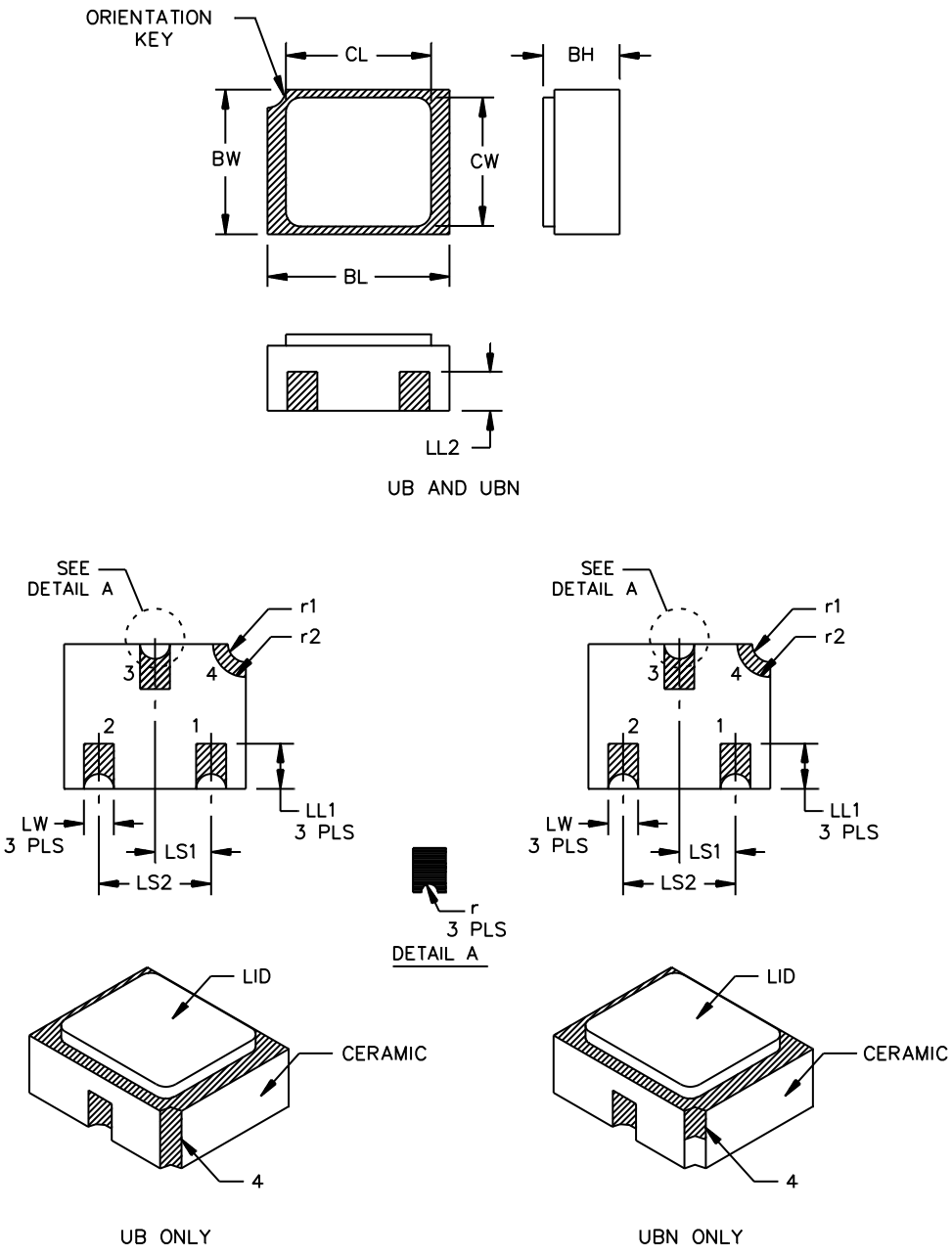



FIGURE 2. Physical dimensions, surface mount 2N3634UB through 2N3637UB (UB and UBN version).



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| Symbol | Dimensions | | | | Note |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| BH | .046 | .056 | 1.17 | 1.42 | |
| BL | .115 | .128 | 2.92 | 3.25 | |
| BW | .085 | .108 | 2.16 | 2.74 | |
| CL | | .128 | | 3.25 | |
| CW | | .108 | | 2.74 | |
| LL1 | .022 | .038 | 0.56 | 0.97 | |
| LL2 | .017 | .035 | 0.43 | 0.89 | |

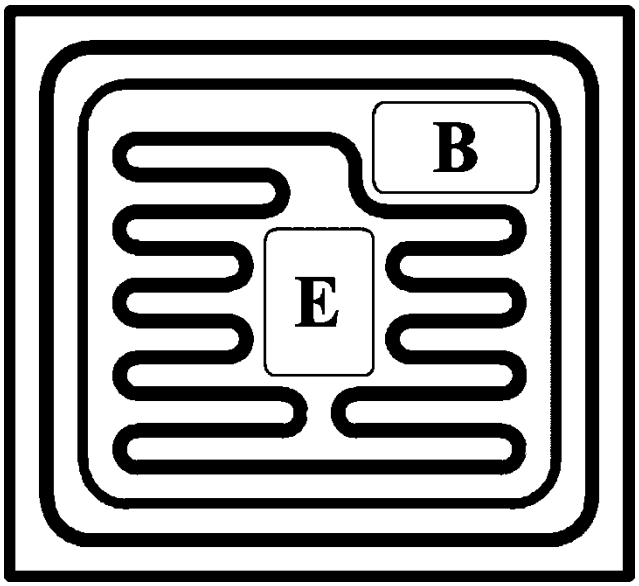
| Symbol | Dimensions | | | | Note |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| $LS_1$ | .036 | .040 | 0.91 | 1.02 | |
| $LS_2$ | .071 | .079 | 1.80 | 2.01 | |
| LW | .016 | .024 | 0.41 | 0.61 | |
| r | | .008 | | .203 | |
| r1 | | .012 | | .305 | |
| r2 | | .022 | | .559 | |
| | | | | | |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Hatched areas on package denote metallized areas.
4. Lid material: Kovar.
5. UB only: Pad 1 = Base, Pad 2 = Emitter, Pad 3 = Collector, Pad 4 = Shielding connected to the metal lid.
6. UBN only: Pad 1 = Base, Pad 2 = Emitter, Pad 3 = Collector, Isolated lid with three pads only.

* FIGURE 2. Physical dimensions, surface mount 2N3634UB through 2N3637UB (UB and UBN version).



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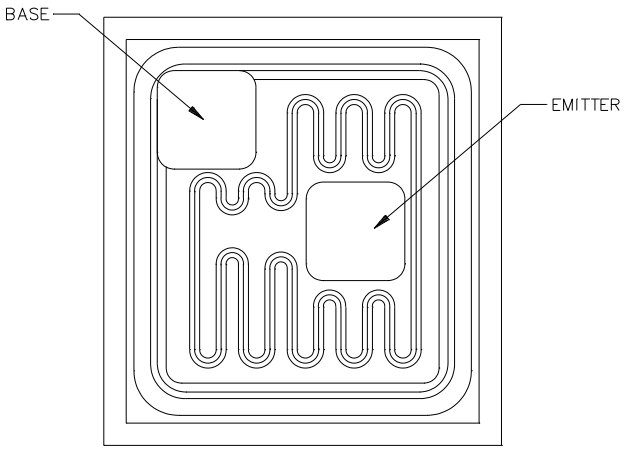



1. Chip size.....…024 x .026 inch ±.002 inch (0.61 x 0.66 mm ±0.051mm).
2. Chip thickness.....…010 ±.0015 inch nominal (0.254 ±0.038 mm).
3. Top metal.....Aluminum 15,000Å minimum, 18,000Å nominal.
4. Back metal.....Gold 3,500Å minimum, 5,000Å nominal.
5. Backside.....…Collector.
6. Bonding pad .....B = .004 x .006 inch, (0.102 mm x 0.152 mm).
   E = .004 x .0055 inch. (0.102 mm x 0.140 mm).

* FIGURE 3. JANHCA and JANKCA die dimensions.



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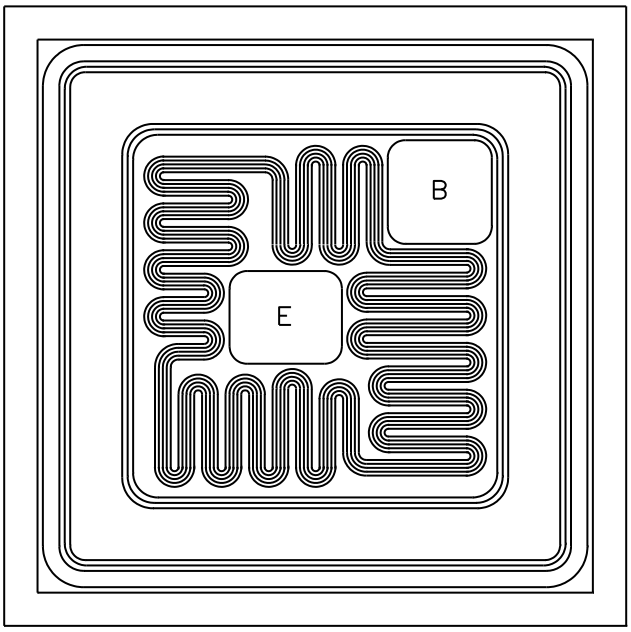



1. Chip size.....….024 x .0259 inch nominal (0.61 x 0.66 mm).
2. Chip thickness.....0079 inch nominal (0.21mm).
3. Top metal.....Aluminum 20,000Å nominal.
4. Back metal.....12K Gold.
5. Backside. .....Collector.

* FIGURE 4. JANHCB and JANKCB die dimensions.



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1. Chip size.....….0.027 x 0.027 inch nominal (0.686 x 0.686 mm).
2. Chip thickness.....…0.008 inch nominal (0.203mm).
3. Top metal.....Aluminum, 24,500Å nominal.
4. Back metal.....Gold, 6,500Å nominal.
5. Backside. .....….Collector.
6. Bonding Pad…….Base = 0.0045 x 0.0045 inch nominal (0.114 x 0.114 mm)
   Emitter = 0.004 x 0.005 inch nominal (0.102 x 0.127 mm)
7. ~~In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.~~

* FIGURE 5. JANHCC and JANKCC die dimensions.



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## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list QML-19500 before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

$R_{\theta JSP(AM)}$ ..... Thermal resistance junction to solder pads (adhesive mount to PCB).
$R_{\theta JSP(IS)}$..... Thermal resistance junction to solder pads (infinite sink mount to PCB).

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1, figure 2, figure 3, figure 4, and figure 5.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Radiation hardness assurance (RHA). Radiation hardness assurance requirements, PIN designators, and test levels shall be as defined in MIL-PRF-19500.



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3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I herein.

* 3.8 Marking. Marking shall be in accordance with MIL-PRF-19500. The radiation hardened designator E, K, U, M, D, P, L, R, F, G, or H shall immediately precede (or replace) the device “2N” identifier (depending upon degree of abbreviation required).

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 JANHC and JANKC qualification. JANHC and JANKC qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.2 Group E qualification. Group E inspection shall be performed for qualification or requalification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot to this revision to maintain qualification.



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4.3 Screening.

4.3.1 Screening of encapsulated devices (JANTX, JANTXV, and JANS levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen | Measurement | |
|---|---|---|
| | JANS level | JANTX and JANTXV levels |
| (1) 3c | Thermal impedance, method 3131 of MIL-STD-750 (see 4.3.4). | Thermal impedance, method 3131 of MIL-STD-750 (see 4.3.4). |
| 9 | $I_{CBO2}$ and $h_{FE5}$ | Not applicable |
| 10 | 24 hours minimum. | 24 hours minimum. |
| 11 | $I_{CBO2}$ and $h_{FE5}$<br>$\Delta I_{CBO2}$ = 100 percent of initial value or 10 nA dc, whichever is greater;<br>$\Delta h_{FE5}$ = ±15 percent of initial value. | $I_{CBO2}$ and $h_{FE5}$ |
| 12 | See 4.3.3, 240 hours minimum. | See 4.3.3. |
| 13 | Subgroups 2 and 3 of table I herein;<br>$\Delta I_{CBO2}$ = 100 percent of initial value or 10 nA dc, whichever is greater.<br>$\Delta h_{FE5}$ = ±15 percent of initial value. | Subgroup 2 of table I herein;<br>$\Delta I_{CBO2}$ = 100 percent of initial value or 10 nA dc, whichever is greater.<br>$\Delta h_{FE5}$ = ±15 percent of initial value. |

(1) Shall be performed anytime after temperature cycling, screen 3a; JANTX and JANTXV levels do not need to be repeated in screening requirements.

4.3.2 Screening of unencapsulated die (JANHC and JANKC). Screening of JANHC and JANKC unencapsulated die shall be in accordance with appendix G of MIL-PRF-19500. The burn-in duration for JANKC level shall follow the JANS requirements; the JANHC level shall follow the JANTX requirements of table E-IV of MIL-PRF-19500.

* 4.3.3 Power burn-in conditions. Power burn-in conditions are as follows: $V_{CB}$ = -10 to -30 V dc. Power shall be applied to achieve $T_J$ = +135°C minimum using a minimum $P_D$ = 75 percent of $P_T$ maximum rated as defined in 1.3. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, $T_J$, and mounting conditions) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

4.3.4 Thermal impedance ($Z_{\theta JX}$ measurements). The $Z_{\theta JX}$ measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{MD}$ (and $V_C$ where appropriate). (See figure 10, figure 11, and figure 12.) See table III, group E, subgroup 4 herein.



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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein. If alternate screening is being performed in accordance with MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of table I, group A, subgroup 1, subgroup 2 inspection only.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) of MIL-PRF-19500 and 4.4.2.1 herein. Delta requirements shall be in accordance with 4.5.4 herein. See 4.4.2.2 for JAN, JANTX, and JANTXV group B testing. Delta requirements for JAN, JANTX, and JANTXV shall be after each step in 4.4.2.2 and shall be in accordance with 4.5.4 herein.

4.4.2.1 Group B inspection, table E-VIA (JANS) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B4 | 1037 | $V_{CB}$ = -10 - -30 V dc; 2,000 cycles. No heat sink or forced-air cooling on devices shall be permitted. |
| B5 | 1027 | $V_{CB}$ = -10 V dc; $P_D \geq$ 100 percent of maximum rated $P_T$ (see 1.3). (NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample.)<br><br>Option 1: 96 hours minimum, sample size in accordance with table E-VIa of MIL-PRF-19500 adjust $T_A$ or $P_D$ to achieve $T_J$ = +275°C minimum.<br><br>Option 2: 216 hours minimum, sample size = 45, c = 0; adjust $T_A$ or $P_D$ to achieve $T_J$ = +225°C minimum. |

4.4.2.2 Group B inspection, table E-VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500. Separate samples may be used for each step. In the event of a lot failure, the resubmission requirements of MIL-PRF-19500 shall apply. In addition, all catastrophic failures during CI shall be analyzed to the extent possible to identify root cause and corrective action. Whenever a failure is identified as wafer lot and /or wafer processing related, the entire wafer lot and related devices assembled from the wafer lot shall be rejected unless an appropriate determined corrective action to eliminate the failures mode has been implemented and the devices from the wafer lot are screened to eliminate the failure mode.

| Step | Method | Condition |
|---|---|---|
| 1 | 1026 | Steady-state life: 1,000 hours minimum, $V_{CB}$ = -10 V dc, power shall be applied to achieve $T_J$ = +150°C minimum using a minimum of $P_D$ = 75 percent of maximum rated $P_T$ as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours. |
| 2 | 1048 | Blocking life, $T_A$ = +150°C, $V_{CB}$ = 80 percent of rated voltage, 48 hours minimum. n = 45 devices, c = 0. |
| 3 | 1032 | High-temperature life (non-operating), t = 340 hours, $T_A$ = +200°C. n = 22, c = 0. |



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4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Shall be chosen from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the test and conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and in 4.4.3.1 (JANS) and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Delta requirements only apply to subgroup C6 and shall be in accordance with 4.5.4 herein.

4.4.3.1 Group C inspection (JANS), table E-VII of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Test condition E; (not applicable for UB and UBN devices). |
| C5 | 3131 | $R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3) and in accordance with thermal impedance curves. |
| C6 | 1026 | Test condition B, 1,000 hours at $V_{CB}$ = -10 V dc; power shall be applied to achieve $T_J$ = +150°C minimum and a minimum of $P_D$ = 75 percent of maximum rated $P_T$ as defined in 1.3. n = 45, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours. |

4.4.3.2 Group C inspection (JAN, JANTX, and JANTXV), table E-VII of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Test condition E; not applicable for UB and UBN devices. |
| C5 | 3131 | $R_{\theta JA}$ and $R_{\theta JC}$ only, (see 1.3 and 4.3.4) and in accordance with thermal impedance curves. |
| C6 | | Not applicable. |

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes table I tests herein for conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for C6 life test may be pulled prior to the application of final lead finish. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group D inspection. Conformance inspection for hardness assured JANS and JANTXV types shall include the group D tests specified in table II herein. These tests shall be performed as required in accordance with MIL-PRF-19500 and method 1019 of MIL-STD-750, for total ionizing dose or method 1017 of MIL-STD-750 for neutron fluence as applicable (see 6.2 herein), except group D, subgroup 2 may be performed separate from other subgroups. Alternate package options may also be substituted for the testing provided there is no adverse effect to the fluence profile.



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4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-IX of MIL-PRF-19500 and as specified herein. Electrical measurements (end-points) shall be in accordance with subgroup 2 of table I and 4.5.4 herein.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Input capacitance. This test shall be conducted in accordance with method 3240 of MIL-STD-750, except the output capacitor shall be omitted.

4.5.3 Noise figure. Noise figure shall be measured using a model 310B Quan Tech Laboratories test set, or equivalent. Conditions shall be as specified in table I herein.

4.5.4 Delta requirements. Delta requirements shall be as specified below:

| Step | Inspection | MIL-STD-750 | | Symbol | Limit | Unit |
|---|---|---|---|---|---|---|
| | | Method | Conditions | | | |
| 1 | Collector-base cutoff current | 3036 | Bias condition D, $V_{CB}$ = -100 V dc | $\Delta I_{CB02}$ (1) | 100 percent of initial value or ±20 nA dc, whichever is greater. | |
| 2 | Forward current transfer ratio | 3076 | $V_{CE}$ = -10 V dc; $I_C$ = -150 mA dc; pulsed see 4.5.1 | $\Delta h_{FE5}$ (1) | ±25 percent change from initial reading. | |

(1) Devices which exceed the table I limits for this test shall not be accepted.

* 4.5.5 Displacement damage characterization. For RHA devices, each supplier shall perform a displacement damage characterization. The characterization shall demonstrate exposure versus data and does not indicate pass or fail criteria. The exposure shall be conducted in accordance with method 1017 of MIL-STD-750. The following details shall apply:

a. Samples may be taken from any wafer of the qualification lot.

b. As a minimum, testing shall be at 2E+12 n/cm$^2$ plus two additional fluence levels chosen by the manufacturer.

c. If the device degrades less than 5 percent of the specification at the highest fluence level, a single data point may be sufficient.

Alternate package options may be substituted for characterization. The displacement damage characterization data shall be made available from the manufacturer.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 2/ | | | | | | |
| Visual and mechanical inspection 3/ | 2071 | | | | | |
| Solderability 3/ 4/ | 2026 | n = 15 leads, c = 0 | | | | |
| Resistance to solvents 3/ 4/ 5/ | 1022 | n = 15 devices, c = 0 | | | | |
| Salt atmosphere (corrosion) | 1041 | n = 6 devices, c = 0<br>(For laser marked devices only) | | | | |
| Temp cycling 3/ 4/ | 1051 | Test condition C, 25 cycles. n = 22 devices, c = 0 | | | | |
| Hermetic seal 4/<br>Fine leak<br>Gross leak | 1071 | n = 22 devices, c = 0 | | | | |
| Electrical measurements 4/ | | Table I, subgroup 2 | | | | |
| Decap internal visual (design verification) 4/ | 2075 | n = 4 device, c = 0 | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance | 3131 | See 4.3.4 | $Z_{\theta JX}$ | | | °C/W |
| Collector to base, cutoff current | 3036 | Bias condition D | | | | |
| 2N3634, 2N3634L, UB, UBN | | $V_{CB}$ = -140 V dc | $I_{CBO1}$ | | -10 | µA dc |
| 2N3635, 2N3635L, UB, UBN | | $V_{CB}$ = -140 V dc | $I_{CBO1}$ | | -10 | µA dc |
| 2N3636, 2N3636L, UB, UBN | | $V_{CB}$ = -175 V dc | $I_{CBO1}$ | | -10 | µA dc |
| 2N3637, 2N3637L, UB, UBN | | $V_{CB}$ = -175 V dc | $I_{CBO1}$ | | -10 | µA dc |
| Emitter to base, cutoff current | 3061 | Bias condition D, $V_{EB}$ = -5 V dc | $I_{EBO1}$ | | -10 | µA dc |
| Breakdown voltage, collector to emitter | 3011 | Bias condition D, $I_C$ = -10 mA dc pulsed (see 4.5.1) | $V_{(BR)CEO}$ | | | |
| 2N3634, 2N3634L, UB, UBN<br>2N3635, 2N3635L, UB, UBN | | | | -140 | | V dc |
| 2N3636, 2N3636L, UB, UBN<br>2N3637, 2N3637L, UB, UBN | | | | -175 | | V dc |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 2 - Continued | | | | | | |
| Collector to base cutoff current | 3036 | Bias condition D, $V_{CB}$ = -100 V dc | $I_{CBO2}$ | | -100 | nA dc |
| * Emitter to base cutoff current | 3061 | Bias condition D, $V_{EB}$ = -3 V dc | $I_{EBO2}$ | | 50 | nA dc |
| Collector to emitter cutoff current | 3041 | Bias condition D, $V_{CE}$ = -100 V dc | $I_{CEO}$ | | -10 | µA dc |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = -10 V dc, $I_C$ = -0.1 mA dc pulsed (see 4.5.1) | $h_{FE1}$ | | | |
| 2N3634, 2N3634L, UB, UBN<br>2N3636, 2N3636L, UB, UBN | | | | 25 | | |
| 2N3635, 2N3635L, UB, UBN<br>2N3637, 2N3637L, UB, UBN | | | | 55 | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = -10 V dc, $I_C$ = -1.0 mA dc pulsed (see 4.5.1) | $h_{FE2}$ | | | |
| 2N3634, 2N3634L, UB, UBN<br>2N3636, 2N3636L, UB, UBN | | | | 45 | | |
| 2N3635, 2N3635L, UB, UBN<br>2N3637, 2N3637L, UB, UBN | | | | 90 | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = -10 V dc, $I_C$ = -10 mA dc pulsed (see 4.5.1) | $h_{FE3}$ | | | |
| 2N3634, 2N3634L, UB, UBN<br>2N3636, 2N3636L, UB, UBN | | | | 50 | | |
| 2N3635, 2N3635L, UB, UBN<br>2N3637, 2N3637L, UB, UBN | | | | 100 | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = -10 V dc, $I_C$ = -50 mA dc pulsed (see 4.5.1) | $h_{FE4}$ | | | |
| 2N3634, 2N3634L, UB, UBN<br>2N3636, 2N3636L, UB, UBN | | | | 50 | 150 | |
| 2N3635, 2N3635L, UB, UBN<br>2N3637, 2N3637L, UB, UBN | | | | 100 | 300 | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = -10 V dc, $I_C$ = -150 mA dc pulsed (see 4.5.1) | $h_{FE5}$ | | | |
| 2N3634, 2N3634L, UB, UBN<br>2N3636, 2N3636L, UB, UBN | | | | 30 | | |
| 2N3635, 2N3635L, UB, UBN<br>2N3637, 2N3637L, UB, UBN | | | | 60 | | |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 2 - Continued | | | | | | |
| Collector to emitter voltage (saturated) | 3071 | $I_C$ = -10 mA dc, $I_B$ = -1 mA dc pulsed (see 4.5.1) | $V_{CE(sat)1}$ | | -0.3 | V dc |
| Collector to emitter voltage (saturated) | 3071 | $I_C$ = -50 mA dc, $I_B$ = -5 mA dc pulsed (see 4.5.1) | $V_{CE(sat)2}$ | | -0.6 | V dc |
| Base-emitter voltage (saturated) | 3066 | Test condition A; $I_C$ = -10 mA dc, $I_B$ = -1.0 mA dc pulsed (see 4.5.1) | $V_{BE(sat)1}$ | | -0.8 | V dc |
| Base-emitter voltage (saturated) | 3066 | Test condition A; $I_C$ = -50 mA dc, $I_B$ = -5 mA dc pulsed (see 4.5.1) | $V_{BE(sat)2}$ | -0.65 | -0.90 | V dc |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_A$ = +150°C | | | | |
| Collector to base cutoff current | 3036 | Bias condition D, $V_{CB}$ = -100 V dc | $I_{CBO3}$ | | -10 | µA dc |
| Low-temperature operation: | | $T_A$ = -55°C | | | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = -10 V dc, $I_C$ = -50 mA dc | $h_{FE6}$ | | | |
| 2N3634, 2N3634L, UB, UBN<br>2N3636, 2N3636L, UB, UBN | | | | 25 | | |
| 2N3635, 2N3635L, UB, UBN<br>2N3637, 2N3637L, UB, UBN | | | | 50 | | |
| Subgroup 4 | | | | | | |
| Small-signal short-circuit forward-current transfer ratio | 3306 | $V_{CE}$ = -30 V dc, $I_C$ = -30 mA dc, f = 100 MHz | $\lvert h_{fe} \rvert$ | | | |
| 2N3634, 2N3634L, UB, UBN<br>2N3636, 2N3636L, UB, UBN | | | | 1.5 | 8.0 | |
| 2N3635, 2N3635L, UB, UBN<br>2N3637, 2N3637L, UB, UBN | | | | 2.0 | 8.5 | |
| Small-signal short-circuit forward current transfer ratio | 3206 | $V_{CE}$ = -10 V dc, $I_C$ = -10 mA dc, f = 1 kHz | $h_{fe}$ | | | |
| 2N3634, 2N3634L, UB, UBN<br>2N3636, 2N3636L, UB, UBN | | | | 40 | 160 | |
| 2N3635, 2N3635L, UB, UBN<br>2N3637, 2N3637L, UB, UBN | | | | 80 | 320 | |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 4 - Continued | | | | | | |
| Small-signal short-circuit input impedance | 3201 | $V_{CE}$ = -10 V dc, $I_C$ = -10 mA dc, f = 1 kHz | $h_{ie}$ | | | |
| 2N3634, 2N3634L, UB, UBN<br>2N3636, 2N3636L, UB, UBN | | | | 100 | 600 | Ω |
| 2N3635, 2N3635L, UB, UBN<br>2N3637, 2N3637L, UB, UBN | | | | 200 | 1,200 | Ω |
| Small signal open circuit reverse voltage transfer ratio | 3211 | $V_{CE}$ = -10 V dc, $I_C$ = -10 mA dc, f = 1 kHz | $h_{re}$ | | $3x10^{-4}$ | |
| Small signal open circuit output admittance | 3216 | $V_{CE}$ = -10 V dc, $I_C$ = -10 mA dc, f = 1 kHz | $h_{oe}$ | | 200 | µs |
| Open circuit output capacitance | 3236 | $V_{CB}$ = -20 V dc, $I_E$ = 0, 100 kHz ≤ f ≤ 1 MHz | $C_{obo}$ | | 10 | pF |
| Input capacitance (output open circuited) | 3240 | $V_{EB}$ = 1 V dc, $I_C$ = 0, 100 kHz ≤ f ≤ 1 MHz | $C_{ibo}$ | | 75 | pF |
| Noise figure | 3246 | $V_{CE}$ = -10 V dc, $I_C$ = -0.5 mA dc, $R_G$ = 1 kΩ (see 4.5.3)<br>f = 100 Hz<br>f = 10 kHz<br>f = 1 kHz | NF | | <br>5<br>3<br>3 | <br>dB<br>dB<br>dB |
| Pulse response | 3251 | Test condition A | | | | |
| Switching parameters | | | | | | |
| Pulse delay time | | See figure 13 | $t_d$ | | 100 | ns |
| Pulse rise time | | See figure 13 | $t_r$ | | 100 | ns |
| Pulse storage time | | See figure 13 | $t_s$ | | 500 | ns |
| Pulse fall time | | See figure 13 | $t_f$ | | 150 | ns |
| * Turn off time ~~$t_{off}$~~ | | $t_s$ & $t_f$ | $t_{off}$ | | 600 | ns |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 5 | | | | | | |
| Safe operating area (continuous dc) TO-39 | 3051 | $T_C$ = + 25°C, t = 1s, 1 cycle (see figures 14 and 15) | | | | |
| Test 1 | | | | | | |
| 2N3634, 2N3634L<br>2N3635, 2N3635L | | $V_{CE}$ = -100 V dc, $I_C$ = -30 mA dc | | | | |
| 2N3636, 2N3636L<br>2N3637, 2N3637L | | $V_{CE}$ = -130 V dc, $I_C$ = -20 mA dc | | | | |
| Test 2 | | $V_{CE}$ = -50 V dc, $I_C$ = -95 mA dc | | | | |
| Test 3 | | $V_{CE}$ = -5 V dc, $I_C$ = -1 A dc | | | | |
| Safe operating area (continuous dc) UB | 3051 | $T_C$ = + 25°C, t = 100 ms, 1 cycle (see figures 16 and 17) | | | | |
| Test 1 | | | | | | |
| 2N3634UB, 2N3635UB,<br>2N3634UBN, 2N3635UBN | | $V_{CE}$ = -85 V dc, $I_C$ = -30 mA dc | | | | |
| 2N3636UB, 2N3637UB<br>2N3636UBN, 2N3637UBN | | $V_{CE}$ = -125 V dc, $I_C$ = -20 mA dc | | | | |
| Test 2 | | $V_{CE}$ = -50 V dc, $I_C$ = -50 mA dc | | | | |
| Test 3 | | $V_{CE}$ = -5 V dc, $I_C$ = -500 mA dc | | | | |
| End-point electrical measurements | | Subgroup 2 of table I. | | | | |
| Subgroups 6 and 7 | | | | | | |
| Not applicable | | | | | | |

1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed table I, subgroup 1, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for JANS devices.
5/ Not required for laser marked devices.



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TABLE II. Group D inspection.

| Inspection 1/ 2/ 3/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 4/ | | | | | | |
| Neutron irradiation | 1017 | Neutron exposure $V_{CES}$ = 0 V | | | | |
| Collector to base, cutoff current | 3036 | Bias condition D | | | | |
| 2N3634, 2N3634L, UB, UBN | | $V_{CB}$ = -140 V dc | $I_{CBO1}$ | | -20 | µA dc |
| 2N3635, 2N3635L, UB, UBN | | $V_{CB}$ = -140 V dc | $I_{CBO1}$ | | -20 | µA dc |
| 2N3636, 2N3636L, UB, UBN | | $V_{CB}$ = -175 V dc | $I_{CBO1}$ | | -20 | µA dc |
| 2N3637, 2N3637L, UB, UBN | | $V_{CB}$ = -175 V dc | $I_{CBO1}$ | | -20 | µA dc |
| Emitter to base, cutoff current | 3061 | Bias condition D, $V_{EB}$ = -5 V dc | $I_{EBO1}$ | | -20 | µA dc |
| Breakdown voltage, collector to emitter | 3011 | Bias condition D, $I_C$ = -10 mA dc pulsed (see 4.5.1) | $V_{(BR)CEO}$ | | | |
| 2N3634, 2N3634L, UB, UBN<br>2N3635, 2N3635L, UB, UBN | | | | -140 | | V dc |
| 2N3636, 2N3636L, UB, UBN<br>2N3637, 2N3637L, UB, UBN | | | | -175 | | V dc |
| Collector to base cutoff current | 3036 | Bias condition D, $V_{CB}$ = -100 V dc | $I_{CBO2}$ | | -200 | nA dc |
| Emitter to base cutoff current | 3061 | Bias condition D, $V_{EB}$ = 3 V dc | $I_{EBO2}$ | | 100 | nA dc |
| Collector to emitter cutoff current | 3041 | Bias condition D, $V_{CE}$ = -100 V dc | $I_{CEO}$ | | -20 | µA dc |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = -10 V dc; $I_C$ = -0.1 mA dc | [$h_{FE1}$] 5/ | | | |
| 2N3634, 2N3636<br>2N3635, 2N3637 | | | | [12.5]<br>[27.5] | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = -10 V dc; $I_C$ = -1.0 mA dc | [$h_{FE2}$] 5/ | | | |
| 2N3634, 2N3636<br>2N3635, 2N3637 | | | | [22.5]<br>[45] | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = -10 V dc; $I_C$ = -10 mA dc | [$h_{FE3}$] 5/ | | | |
| 2N3634, 2N3636<br>2N3635, 2N3637 | | | | [25]<br>[50] | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = -10 V dc; $I_C$ = -50 mA dc | [$h_{FE4}$] 5/ | | | |
| 2N3634, 2N3636<br>2N3635, 2N3637 | | | | [25]<br>[50] | 150<br>300 | |

See footnotes at end of table.



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TABLE II. Group D inspection - Continued.

| | Inspection 1/ 2/ 3/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| | Subgroup 1 - Continued. | | | | | | |
| | Forward-current transfer ratio | 3076 | $V_{CE}$ = -10 V dc; $I_C$ = -150 mA dc | [$h_{FE5}$] 5/ | | | |
| | 2N3634, 2N3636<br>2N3635, 2N3637 | | | | [15]<br>[30] | | |
| | Collector-emitter saturation voltage | 3071 | $I_C$ = -10 mA dc; $I_B$ = -1.0 mA dc | $V_{CE(sat)1}$ | | -.35 | V dc |
| | Collector-emitter saturation voltage | 3071 | $I_C$ = -50 mA dc; $I_B$ = -5.0 mA dc | $V_{CE(sat)2}$ | | -.69 | V dc |
| | Base-emitter voltage (saturated) | 3066 | Test condition A; $I_C$ = -10 mA dc, $I_B$ = -1.0 mA dc pulsed (see 4.5.1) | $V_{BE(sat)1}$ | | -.92 | V dc |
| | Base-emitter voltage (saturated) | 3066 | Test condition A; $I_C$ = -50 mA dc, $I_B$ = -5 mA dc pulsed (see 4.5.1) | $V_{BE(sat)2}$ | -.65 | -1.04 | V dc |
| | Subgroup 2 | | | | | | |
| * | Total dose irradiation | 1019 | Gamma exposure<br>Condition A for High dose rate,<br>Or Condition D for low dose rate | | | | |
| | 2N3634, 2N3636<br>2N3635, 2N3637 | | $V_{CES}$ = -112 V<br>$V_{CES}$ = -140 V | | | | |
| | Collector to base, cutoff current | 3036 | Bias condition D | | | | |
| | 2N3634, 2N3634L, UB, UBN | | $V_{CB}$ = -140 V dc | $I_{CBO1}$ | | -20 | µA dc |
| | 2N3635, 2N3635L, UB, UBN | | $V_{CB}$ = -140 V dc | $I_{CBO1}$ | | -20 | µA dc |
| | 2N3636, 2N3636L, UB, UBN | | $V_{CB}$ = -175 V dc | $I_{CBO1}$ | | -20 | µA dc |
| | 2N3637, 2N3637L, UB, UBN | | $V_{CB}$ = -175 V dc | $I_{CBO1}$ | | -20 | µA dc |
| * | Emitter to base, cutoff current | 3061 | Bias condition D, $V_{EB}$ = -5 V dc | $I_{EBO1}$ | | 20 | µA dc |
| | Breakdown voltage, collector to emitter | 3011 | Bias condition D, $I_C$ = -10 mA dc pulsed (see 4.5.1) | $V_{(BR)CEO}$ | | | |
| | 2N3634, 2N3634L, UB, UBN<br>2N3635, 2N3635L, UB, UBN | | | | -140 | | V dc |
| | 2N3636, 2N3636L, UB, UBN<br>2N3637, 2N3637L, UB, UBN | | | | -175 | | V dc |
| | Collector to base cutoff current | 3036 | Bias condition D, $V_{CB}$ = -100 V dc | $I_{CBO2}$ | | -200 | nA dc |
| * | Emitter to base cutoff current | 3061 | Bias condition D, $V_{EB}$ = -3 V dc | $I_{EBO2}$ | | 100 | nA dc |
| | Collector to emitter cutoff current | 3041 | Bias condition D, $V_{CE}$ = -100 V dc | $I_{CEO}$ | | -20 | µA dc |

See footnotes at end of table.



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TABLE II. Group D inspection - Continued.

| Inspection 1/ 2/ 3/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 2 - Continued. | | | | | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = -10 V dc; $I_C$ = -0.1 mA dc | [$h_{FE1}$] 5/ | | | |
| 2N3634, 2N3636<br>2N3635, 2N3637 | | | | [12.5]<br>[27.5] | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = -10 V dc; $I_C$ = -1.0 mA dc | [$h_{FE2}$] 5/ | | | |
| 2N3634, 2N3636<br>2N3635, 2N3637 | | | | [22.5]<br>[45] | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = -10 V dc; $I_C$ = -10 mA dc | [$h_{FE3}$] 5/ | | | |
| 2N3634, 2N3636<br>2N3635, 2N3637 | | | | [25]<br>[50] | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = -10 V dc; $I_C$ = -50 mA dc | [$h_{FE4}$] 5/ | | | |
| 2N3634, 2N3636<br>2N3635, 2N3637 | | | | [25]<br>[50] | 150<br>300 | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = -10 V dc; $I_C$ = -150 mA dc | [$h_{FE5}$] 5/ | | | |
| 2N3634, 2N3636<br>2N3635, 2N3637 | | | | [15]<br>[30] | | |
| Collector-emitter saturation voltage | 3071 | $I_C$ = -10 mA dc; $I_B$ = -1.0 mA dc; | $V_{CE(sat)1}$ | | -.35 | V dc |
| Collector-emitter saturation voltage | 3071 | $I_C$ = -50 mA dc; $I_B$ = -5.0 mA dc; | $V_{CE(sat)2}$ | | -.69 | V dc |
| Base-emitter voltage (saturated) | 3066 | Test condition A; $I_C$ = -10 mA dc, $I_B$ = -1.0 mA dc pulsed (see 4.5.1) | $V_{BE(sat)1}$ | | -.92 | V dc |
| Base-emitter voltage (saturated) | 3066 | Test condition A; $I_C$ = -50 mA dc, $I_B$ = -5 mA dc pulsed (see 4.5.1) | $V_{BE(sat)2}$ | .65 | -1.04 | V dc |

1/ Tests to be performed on all devices receiving radiation exposure.
2/ For sampling plan, see MIL-PRF-19500.
3/ Electrical characteristics apply to device types unless otherwise noted.
4/ Subgroup 1 is an optional test and must be specified on the contract when required.
5/ See method 1019 of MIL-STD-750 for how to determine [$h_{FR}$] by first calculating the delta ($1/h_{FE}$) from the pre- and post-radiation $h_{FE}$. Notice the [$h_{FE}$] is not the same as $h_{FE}$ and cannot be measured directly. The [$h_{FE}$] value can never exceed the pre-radiation minimum $h_{FE}$ that it is based upon.



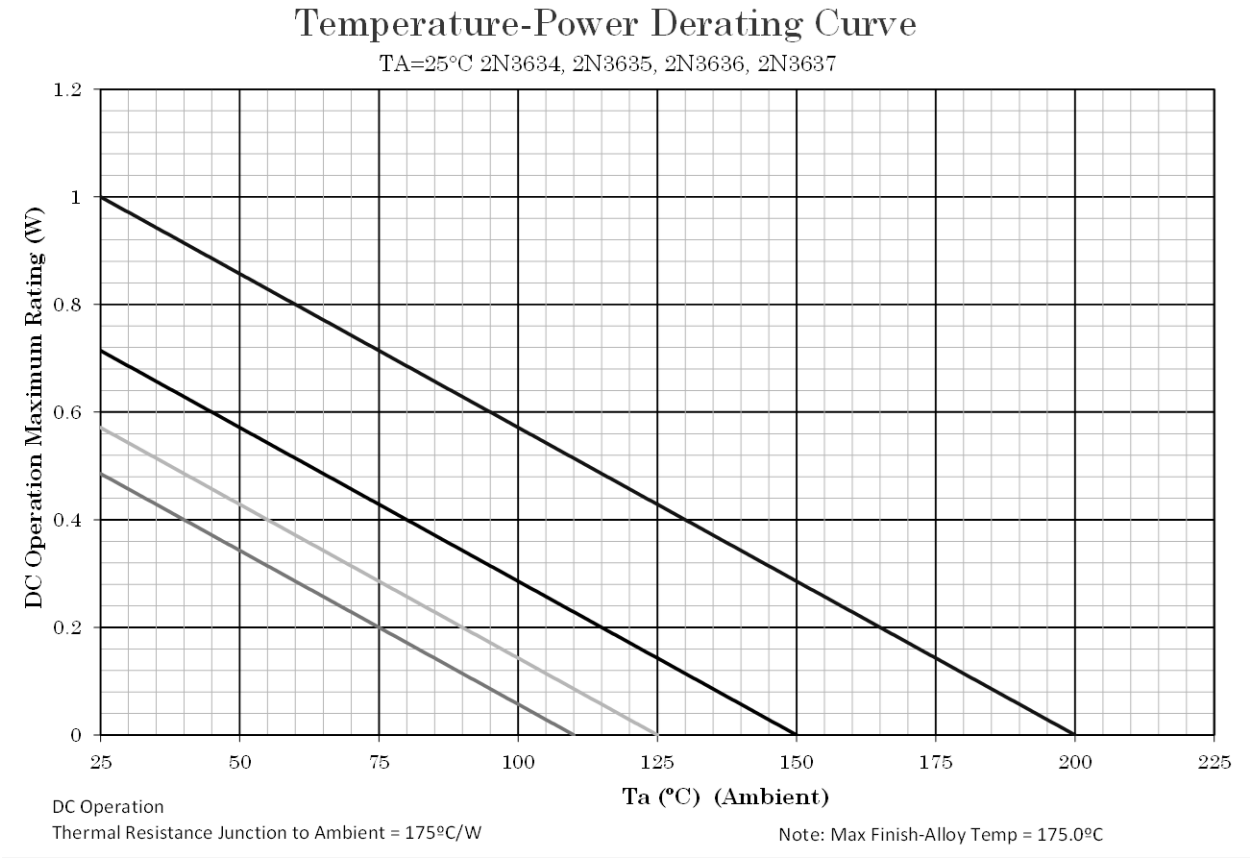
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TABLE III. Group E inspection (all quality levels) - for qualification only.

| Inspection | MIL-STD-750 | | Qualification |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices<br>c = 0 |
| Temperature cycling (air to air) | 1051 | Test condition C, 500 cycles | |
| Hermetic seal | 1071 | | |
| Fine leak<br>Gross leak | | | |
| Electrical measurements | | See table I, subgroup 2 and 4.5.4 herein. | |
| Subgroup 2 | | | 45 devices<br>c = 0 |
| Intermittent life | 1037 | $V_{CB}$ = -10 V dc, 6,000 cycles. Adjust device current, or power, to achieve a minimum $\Delta T_J$ of +100°C. | |
| Electrical measurements | | See table I of subgroup 2 and 4.5.4 herein. | |
| Subgroup 4 | | | 15 devices,<br>c = 0 |
| Thermal resistance | 3131 | $R_{\theta JSP}$ can be calculated but shall be measured once in the same package with a similar die size to confirm calculations (may apply to multiple specification sheets). | |
| Thermal impedance curves | | See MIL-PRF-19500, table E-IX, group E, subgroup 4. | Sample size<br>N/A |
| Subgroup 5 | | | |
| Not applicable | | | |
| Subgroup 8 | | | 45 devices<br>c = 0 |
| Reverse stability | 1033 | Condition B. | |
| * Subgroup 12 | | | |
| Neutron Irradiation | 1017 | See 4.5.5 | |



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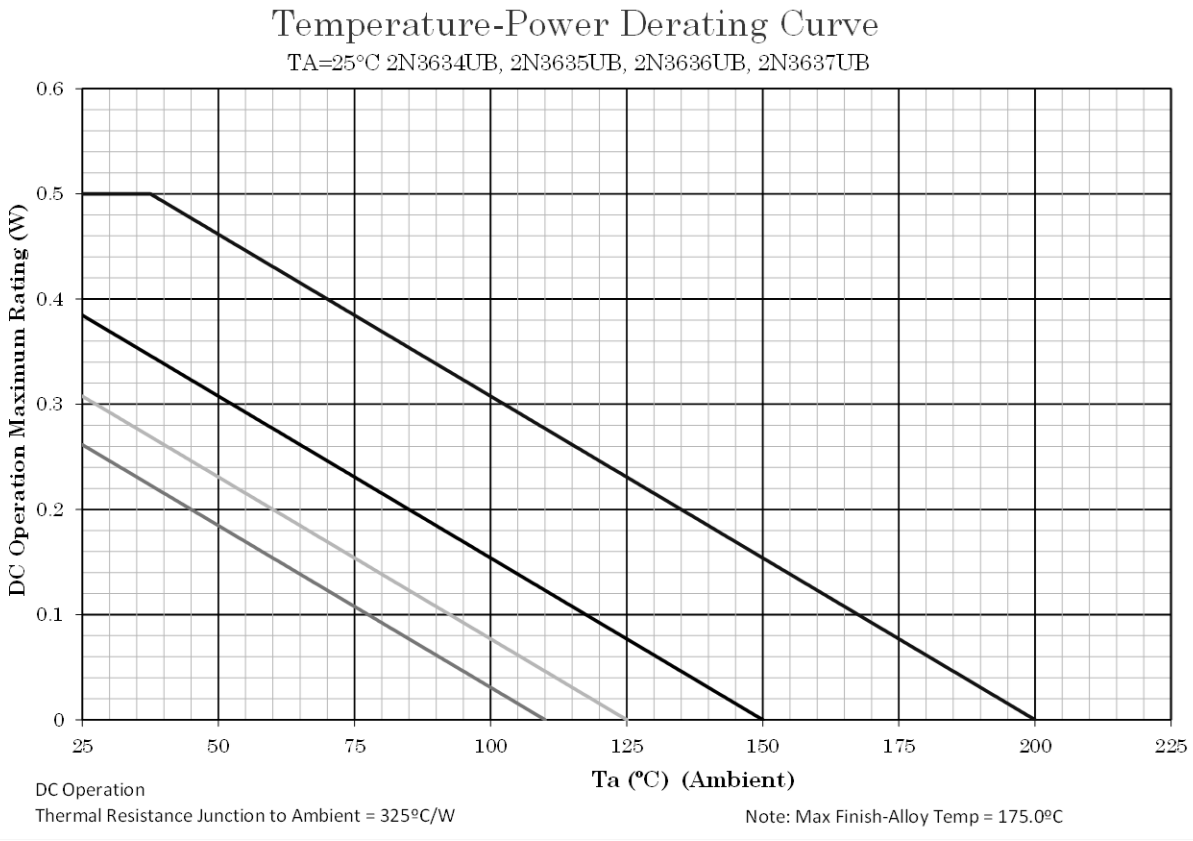

NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J$ ≤ 200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J$ ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J$ ≤ 125°C, and 110°C to show power rating where most users want to limit $T_J$ in their application.

FIGURE 6. Temperature-power derating for ($R_{\theta JA}$), base case mount (TO-5 and TO-39).



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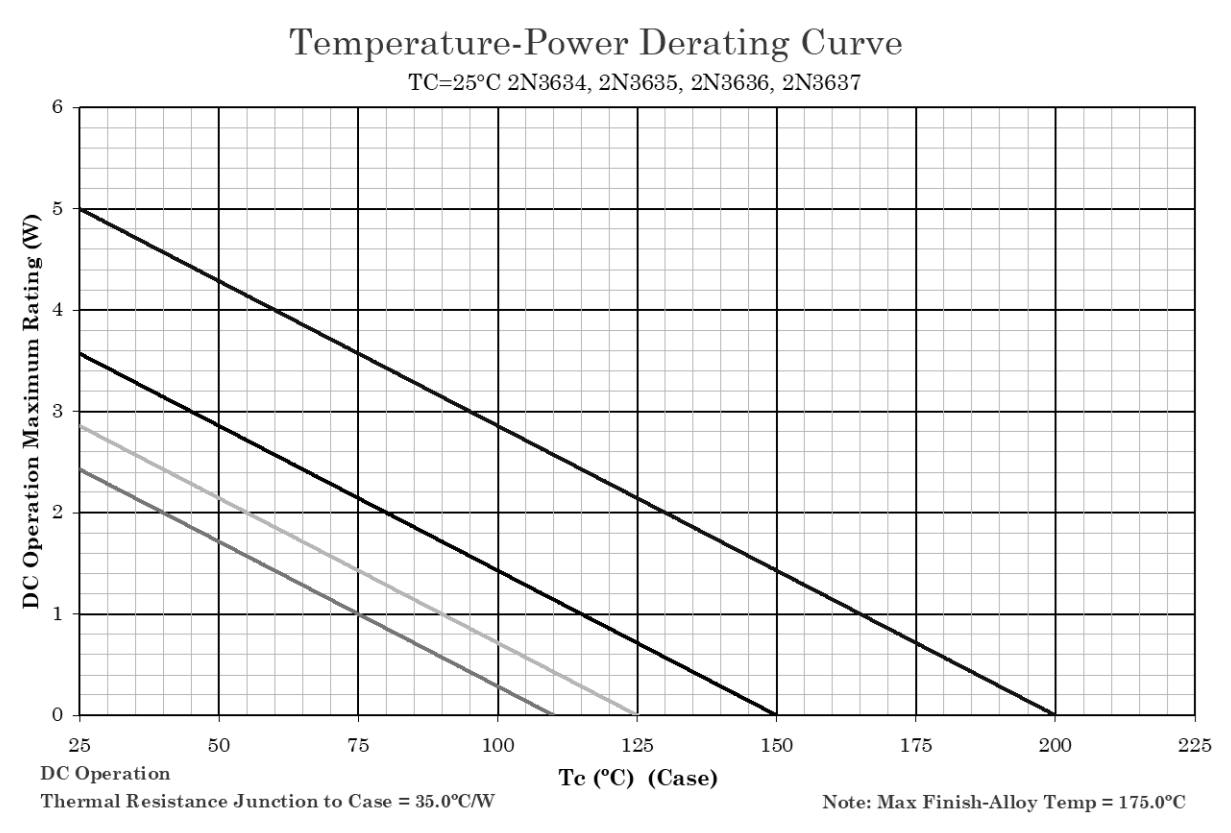

NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J$ ≤ 200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J$ ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J$ ≤ 125°C, and 110°C to show power rating where most users want to limit $T_J$ in their application.

FIGURE 7. Temperature-power derating for ($R_{\theta JA}$) (UB AND UBN).



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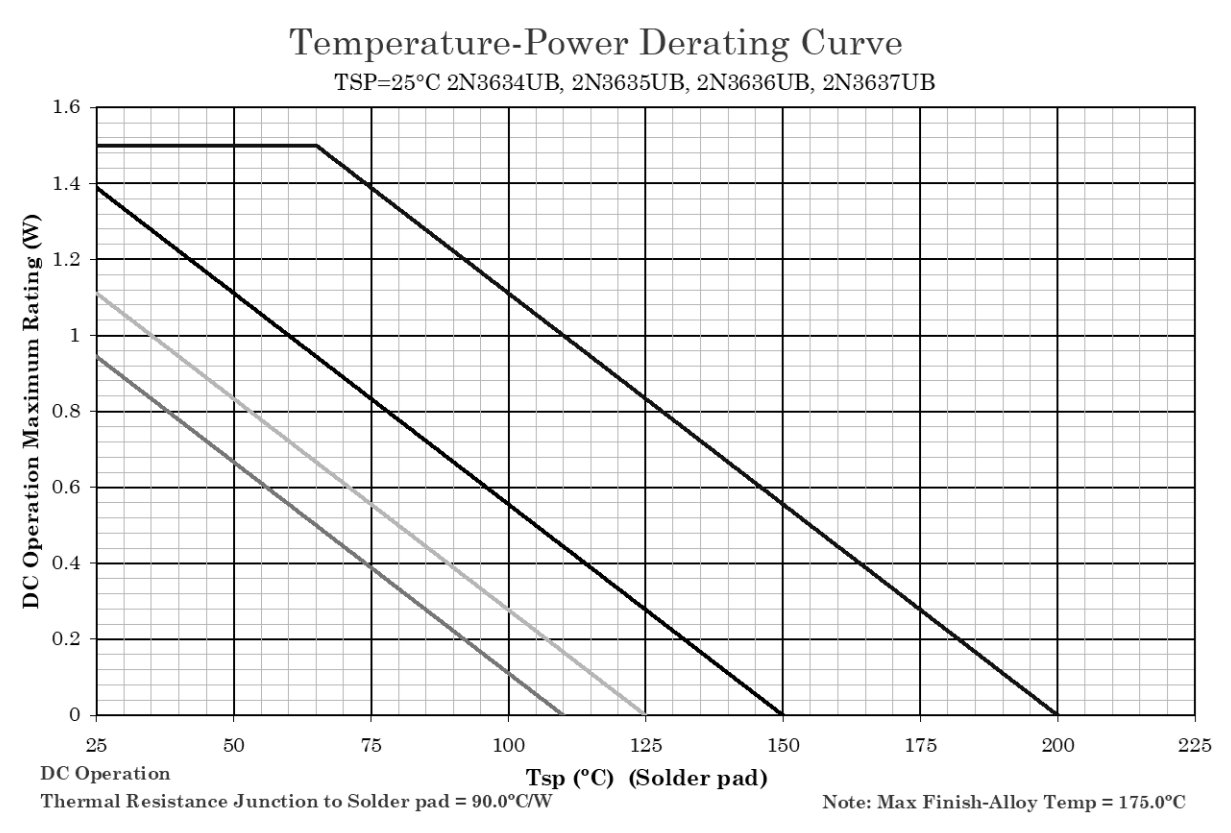

NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J$ ≤ 200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J$ ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J$ ≤ 125°C, and 110°C to show power rating where most users want to limit $T_J$ in their application.

FIGURE 8. Temperature-power derating for ($R_{\theta JC}$), base case mount (TO-5 and TO-39 Kovar).



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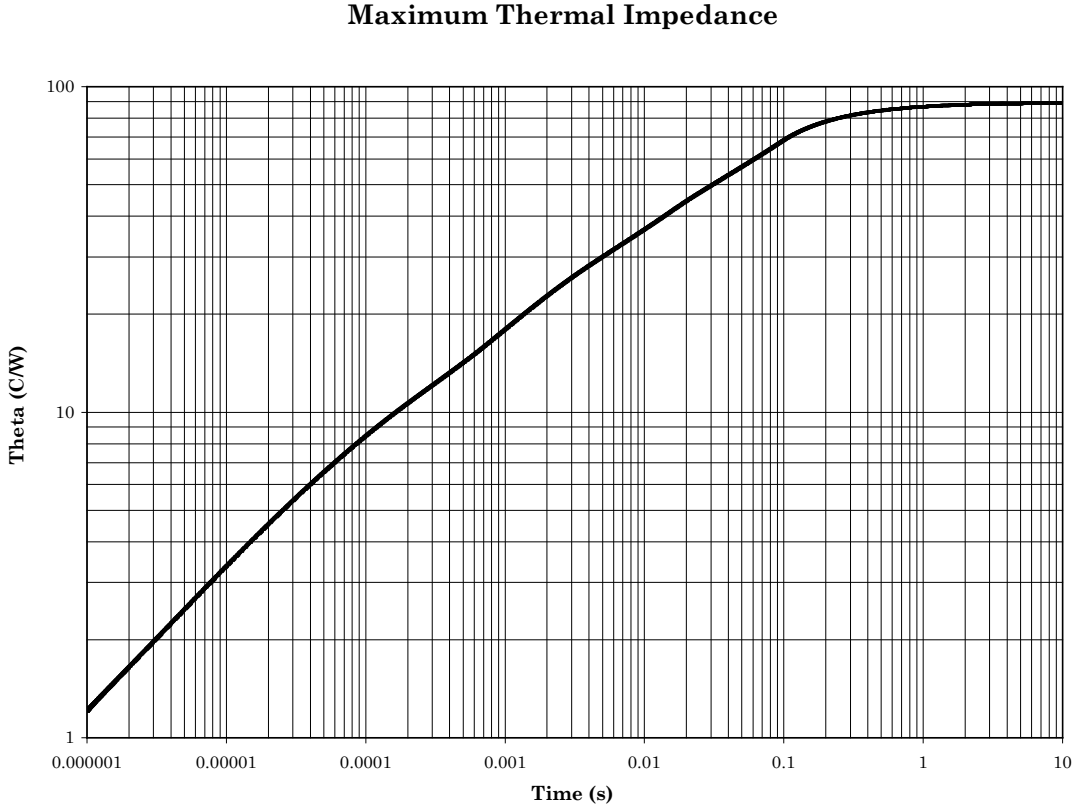

NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J$ ≤ 200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J$ ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J$ ≤ 125°C, and 110°C to show power rating where most users want to limit $T_J$ in their application.

FIGURE 9. Temperature-power derating for ($R_{\theta JSP}$), base case mount (UB AND UBN).



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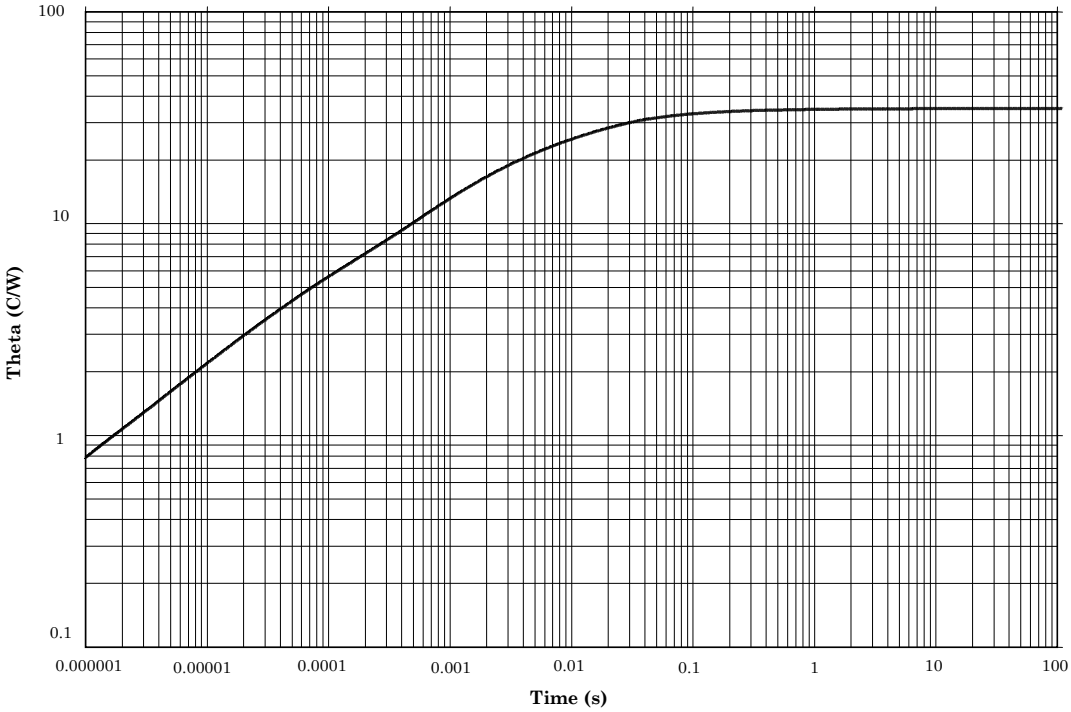

$T_{SP}$ = +25°C, thermal resistance $R_{\theta JSP}$ = 90°C/W (solder mounted to heavy copper clad PCB).

FIGURE 10. Thermal impedance graph ($R_{\theta JSP}$) for 2N3634UB through 2N3637UB (UB AND UBN).



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**Maximum Thermal Impedance**



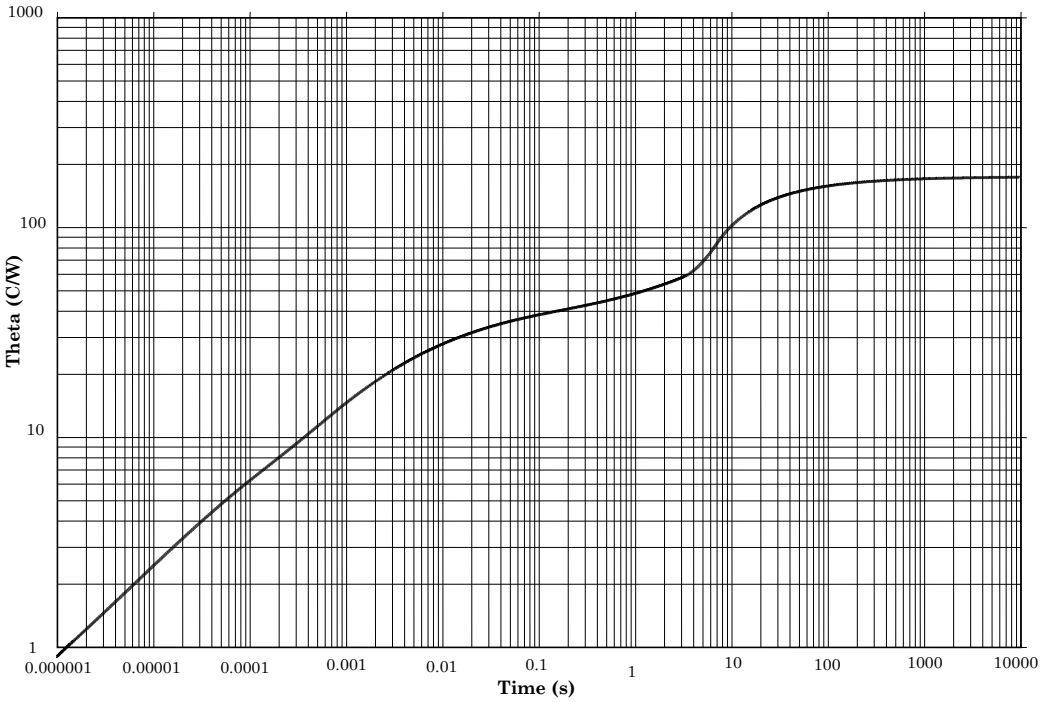

$T_C$ = +25°C, thermal resistance $R_{\theta JC}$ = 35°C/W (ambient case mount).

FIGURE 11. Thermal impedance graph ($R_{\theta JC}$) for 2N3634 through 2N3637 and 2N3634L through 2N3637L (TO-5 and TO-39 Kovar).



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**Maximum Thermal Impedance**

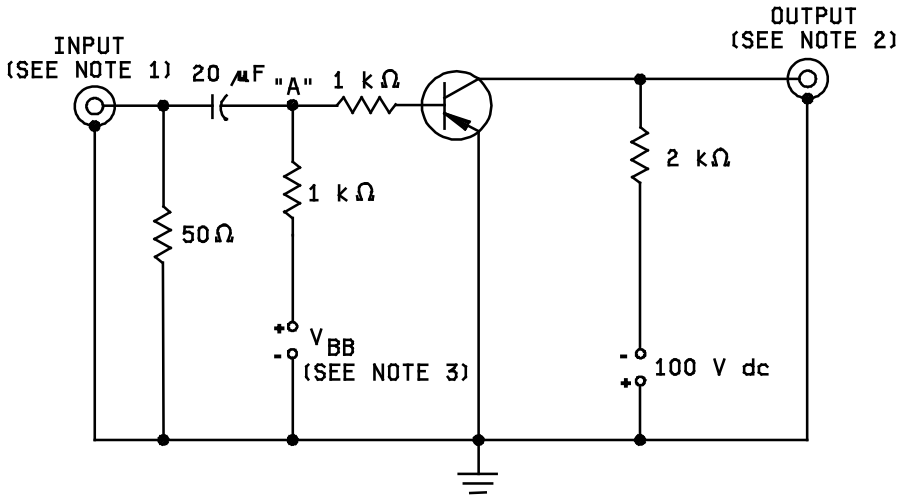



$T_A$ = +25°C, thermal resistance $R_{\theta JA}$ = 175°C/W (ambient free air cooled).

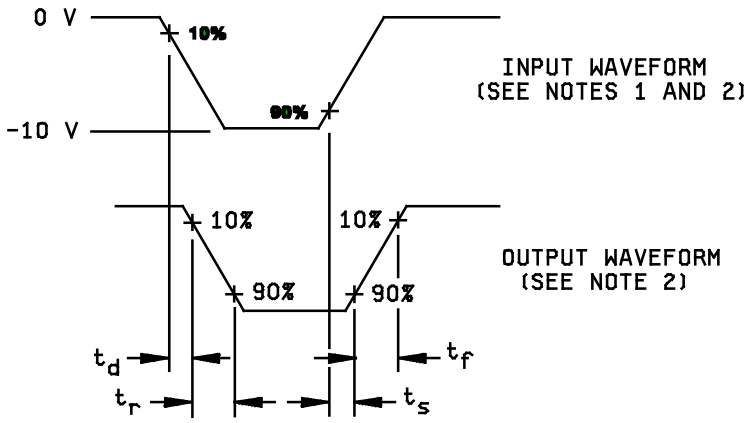
FIGURE 12. Thermal impedance graph ($R_{\theta JA}$) for 2N3634 through 2N3637 and 2N3634L through 2N3637L (TO-5 and TO-39).



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TEST CIRCUIT

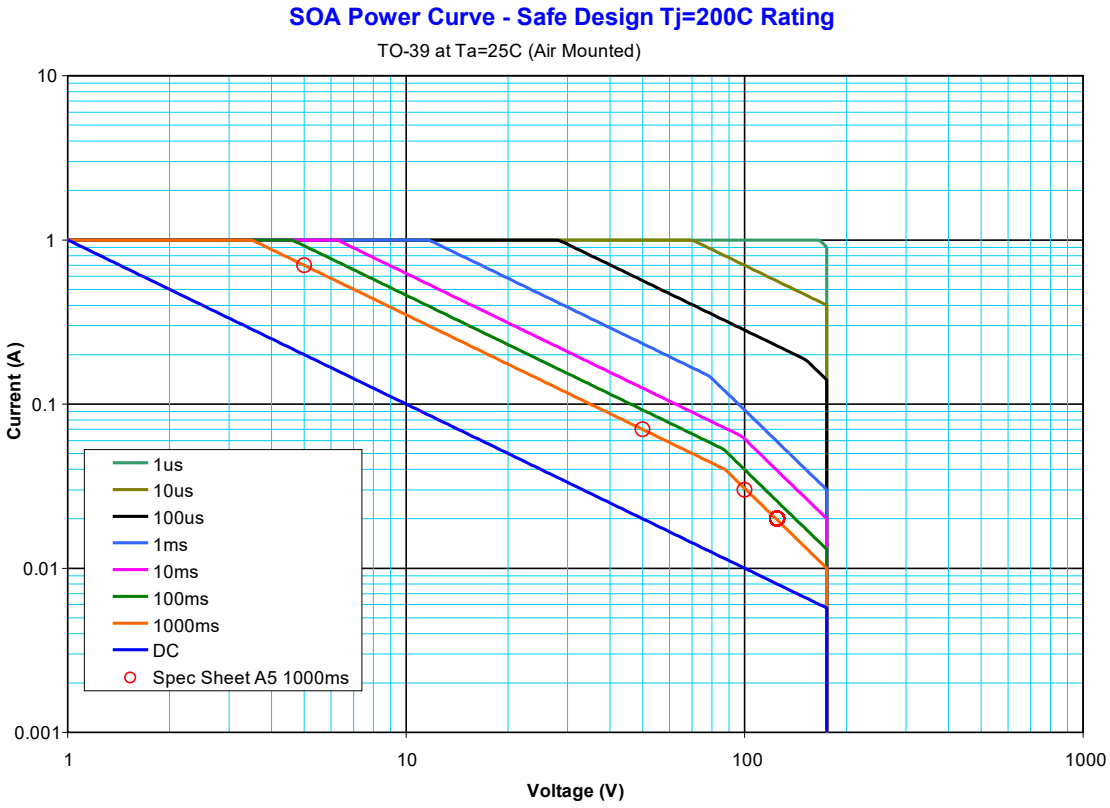



NOTES:
1. The input waveform is supplied by a pulse generator with the following characteristics: $t_r \leq 15$ ns, $t_f \leq 15$ ns, $Z_{OUT} = 50\ \Omega$, PW = 20 µs, duty cycle ≤ 2 percent.
2. Output waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 15$ ns, $R_{in} \geq 10\ M\Omega$, $C_{in} \leq 11.5$ pF.
3. $V_{BB}$ = 4.0 V dc for $t_{on}$, 4.1 V dc for $t_{off}$ at point “A”.
4. Resistors shall be noninductive types.
5. The dc power supplies may require additional by-passing in order to minimize ringing.

FIGURE 13. Pulse response test circuit.



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2N3637 $T_A$ = +25°C. A5 limits at 1,000 ms.

* $V_{CE}$ = -5V at 0.7A,

* $V_{CE}$ = -50V at 0.07A,

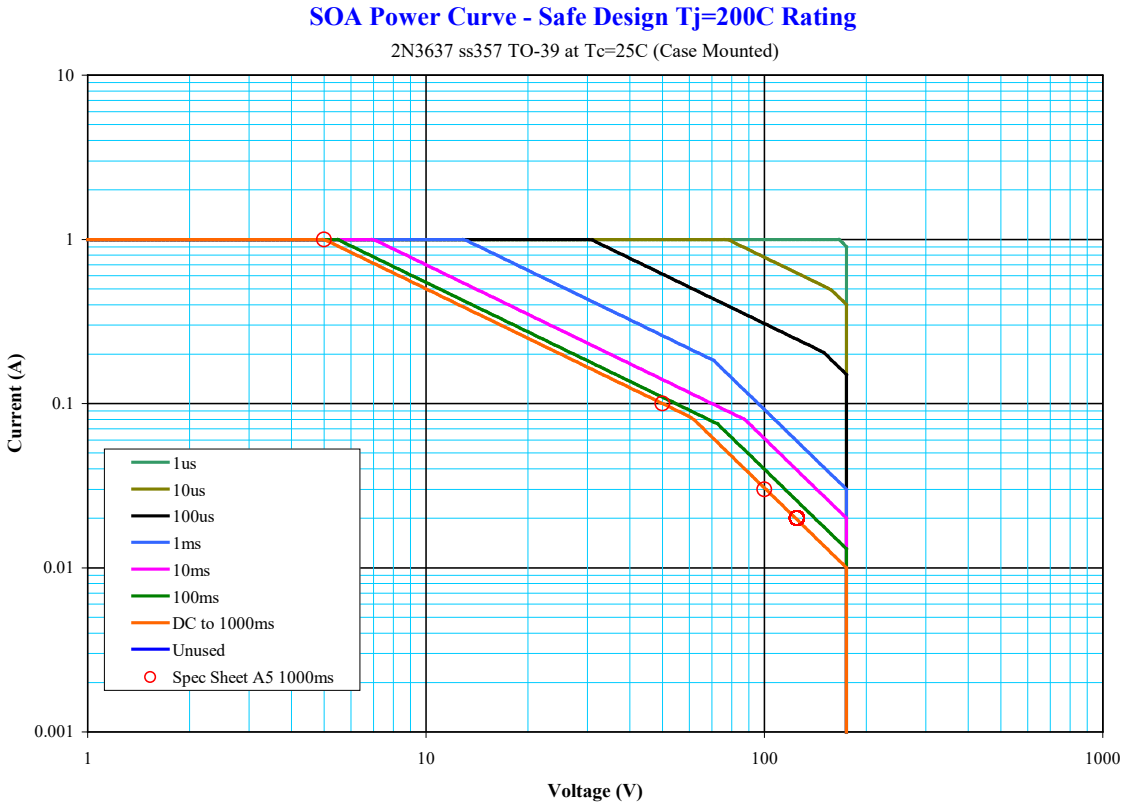
* $V_{CE}$ = -100V at 0.03A,

* $V_{CE}$ = -125V at 0.02A.

FIGURE 14. Maximum safe operating area graph (continuous dc).



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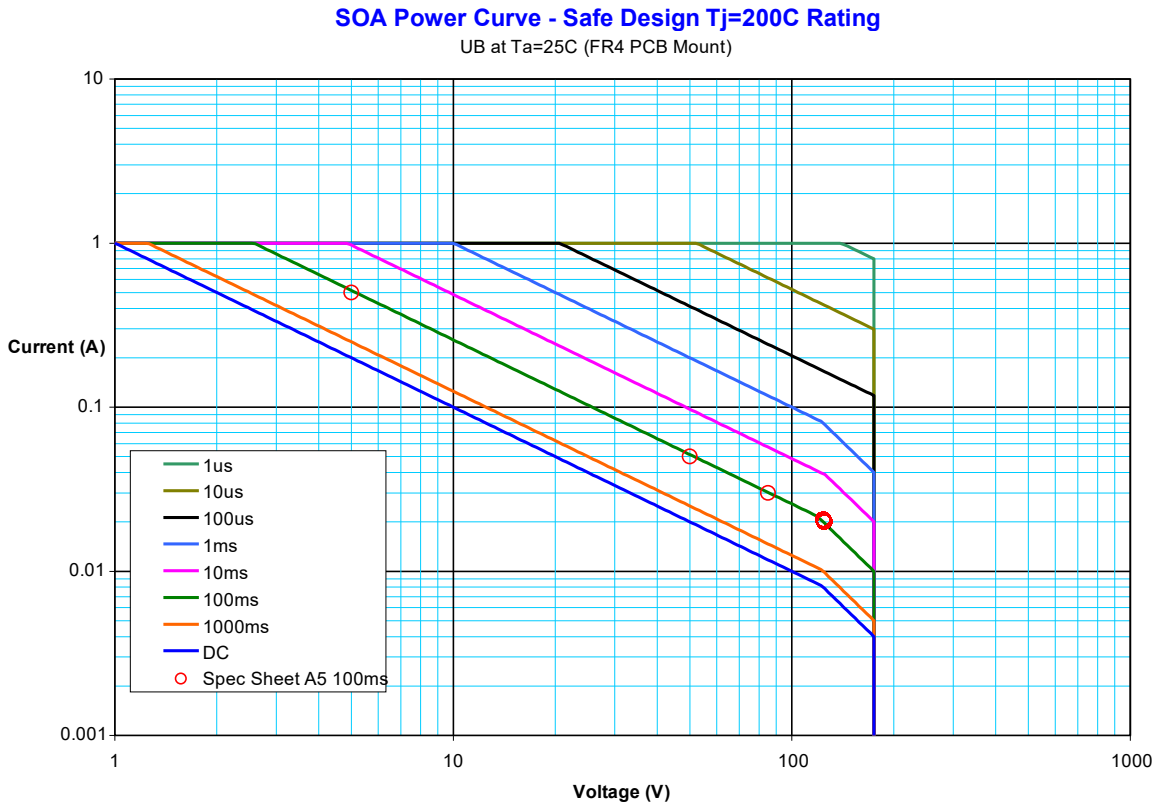



2N3637 $T_C$ = +25°C. A5 limits at 1,000 ms.
$V_{CE}$ = -5V at 1A,
$V_{CE}$ = -50V at 0.095A,
$V_{CE}$ = -100V at 0.03A,
$V_{CE}$ = -130V at 0.02A.

FIGURE 15. Maximum safe operating area graph (continuous dc).



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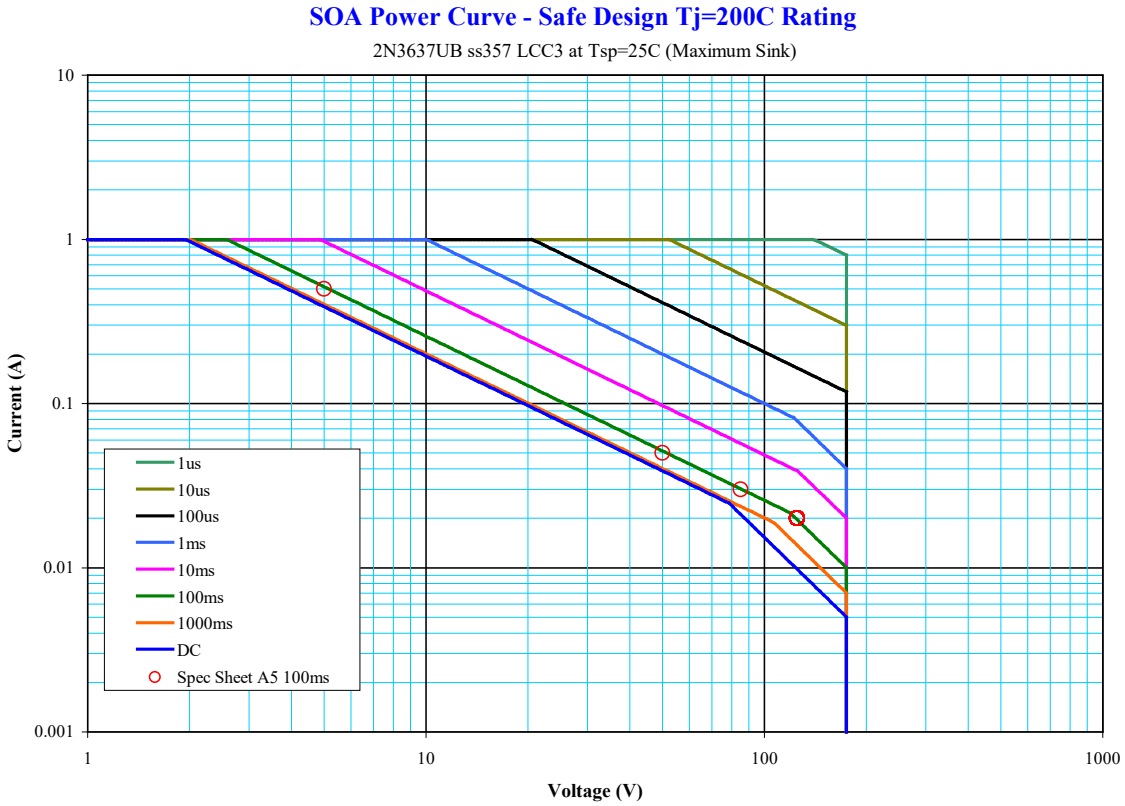



2N3637UB $T_A$ = +25°C. A5 limits at 100 ms.
* $V_{CE}$ = -5V at 0.5A,
* $V_{CE}$ = -50V at 0.05A,
* $V_{CE}$ = -85V at 0.03A,
* $V_{CE}$ = -125V at 0.02A.

FIGURE 16. Maximum safe operating area graph for UB AND UBN (continuous dc).



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2N3637UB $T_{SP}$ = +25°C. A5 limits at 100 ms.
* $V_{CE}$ = -5V at 0.5A,
* $V_{CE}$ = -50V at 0.05A,
* $V_{CE}$ = -85V at 0.03A,
* $V_{CE}$ = -125V at 0.02A.

FIGURE 17. Maximum safe operating area graph for UB AND UBN (continuous dc).



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete Part or Identifying Number (PIN), see 1.5 and 6.6.

e. For die acquisition, the JANHC or JANKC letter version shall be specified (see figures 3, 4, and 5).

f. Surface mount designation if applicable.

g. For acquisition of RHA designed devices, table II, subgroup 1 testing of group D is optional. If subgroup 1 testing is desired, it must be specified in the contract.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

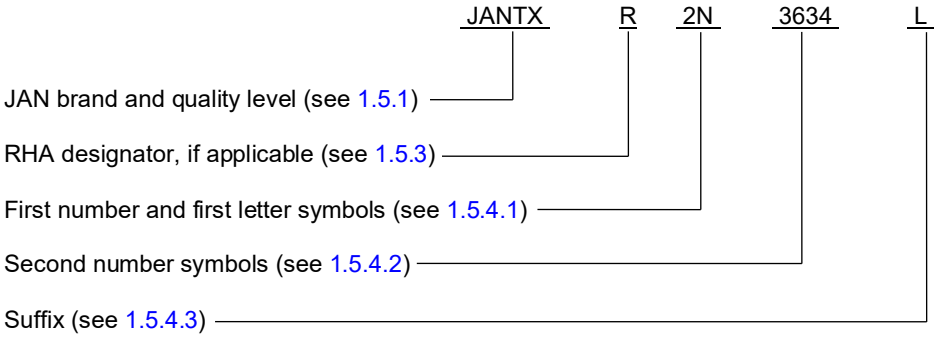
6.4 Supersession information. Devices covered by this specification supersede the manufacturers' and users' Part or Identifying Number (PIN). The term Part or Identifying Number (PIN) is equivalent to the term part number which was previously used in this specification. This information in no way implies that manufacturers' PINs are suitable as a substitute for the military PIN.



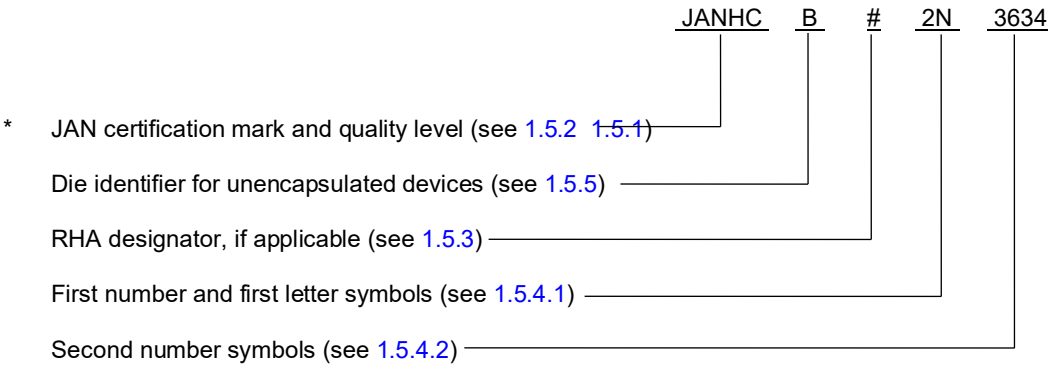
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6.5 PIN construction example.

6.5.1 PIN construction example for encapsulated devices. The PINs for encapsulated devices are construction using the following form.




6.5.2 Un-encapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.






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6.6 List of PINs.

6.6.1 List of PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

| PINs for devices of the base quality level | PINs for devices of the "TX" quality level | PINs for devices of the "TXV" quality level (1) | PINs for devices of the "S" quality level (1) |
|---|---|---|---|
| JAN2N3634 | JANTX2N3634 | JANTXV#2N3634 | JANS#2N3634 |
| JAN2N3634L | JANTX2N3634L | JANTXV#2N3634L | JANS#2N3634L |
| JAN2N3634UB | JANTX2N3634UB | JANTXV#2N3634UB | JANS#2N3634UB |
| JAN2N3634UBN | JANTX2N3634UBN | JANTXV#2N3634UBN | JANS#2N3634UBN |
| JAN2N3635 | JANTX2N3635 | JANTXV#2N3635 | JANS#2N3635 |
| JAN2N3635L | JANTX2N3635L | JANTXV#2N3635L | JANS#2N3635L |
| JAN2N3635UB | JANTX2N3635UB | JANTXV#2N3635UB | JANS#2N3635UB |
| JAN2N3635UBN | JANTX2N3635UBN | JANTXV#2N3635UBN | JANS#2N3635UBN |
| JAN2N3636 | JANTX2N3636 | JANTXV#2N3636 | JANS#2N3636 |
| JAN2N3636L | JANTX2N3636L | JANTXV#2N3636L | JANS#2N3636L |
| JAN2N3636UB | JANTX2N3636UB | JANTXV#2N3636UB | JANS#2N3636UB |
| JAN2N3636UBN | JANTX2N3636UBN | JANTXV#2N3636UBN | JANS#2N3636UBN |
| JAN2N3637 | JANTX2N3637 | JANTXV#2N3637 | JANS#2N3637 |
| JAN2N3637L | JANTX2N3637L | JANTXV#2N3637L | JANS#2N3637L |
| JAN2N3637UB | JANTX2N3637UB | JANTXV#2N3637UB | JANS#2N3637UB |
| JAN2N3637UBN | JANTX2N3637UBN | JANTXV#2N3637UBN | JANS#2N3637UBN |

* (1) The number sign (#) represent one of eleven RHA designators available (E, K, U, M, D, P, L, R, F, G, or H) if desired. Remove for no RHA.



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6.6.2 List of PINs for unencapsulated devices. The following is a list of possible PINs available on this specification sheet. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCA2N3634) will be identified on the QML.

| JANHC and JANKC ordering information (1) | | | |
|---|---|---|---|
| PIN | Manufacturer | | |
| | 43611 | 31471 | 34156 |
| 2N3634 | JANHCA#2N3634<br>JANKCA#2N3634 | JANHCB#2N3634<br>JANKCB#2N3634 | JANHCC#2N3634<br>JANKCC#2N3634 |
| 2N3635 | JANHCA#2N3635<br>JANKCA#2N3635 | JANHCB#2N3635<br>JANKCB#2N3635 | JANHCC#2N3635<br>JANKCC#2N3635 |
| 2N3636 | JANHCA#2N3636<br>JANKCA#2N3636 | JANHCB#2N3636<br>JANKCB#2N3636 | JANHCC#2N3636<br>JANKCC#2N3636 |
| 2N3637 | JANHCA#2N3637<br>JANKCA#2N3637 | JANHCB#2N3637<br>JANKCB#2N3637 | JANHCC#2N3637<br>JANKCC#2N3637 |

* (1) The number sign (#) represent one of eleven RHA designators available (E, K, U, M, D, P, L, R, F, G, or H) if desired. Remove for no RHA.

* 6.7 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, MC, SH
Air Force - 19, 71

Preparing activity:
DLA - CC

(Project 5961-2024-029)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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